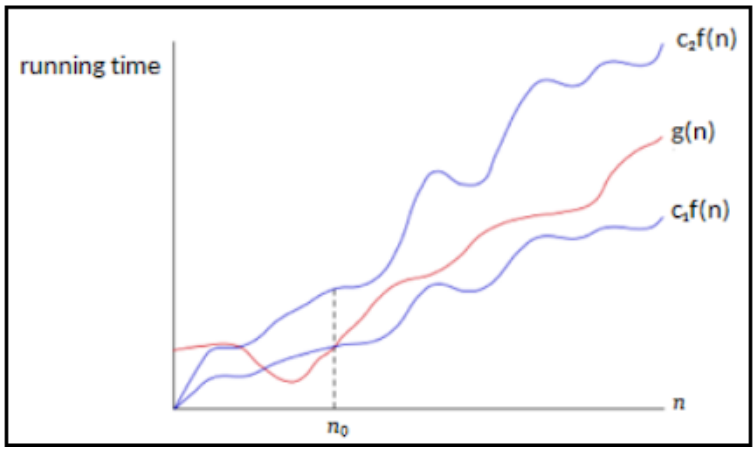
NOTATII ASIMPTOTICE

Θ

- aceeași rata de creștere
- f ∈ Θ(f)

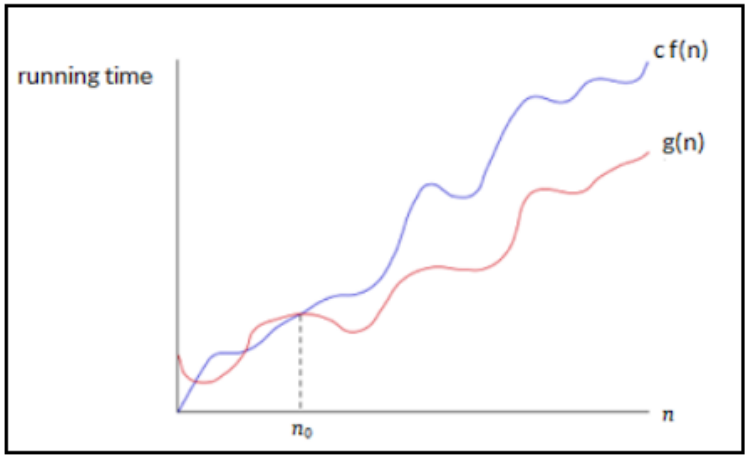
$$\Theta(f(n)) = \{g : \mathbb{R}^+ \to \mathbb{R}^+ \mid \begin{matrix}\exists c_1, c_2 \in \mathbb{R}^+ \\ \exists n_0 \in \mathbb{N}\end{matrix} \; such\ that \; \forall\, n \geq n_0,\ \ c_1 f(n) \leq g(n) \leq c_2 f(n)\}$$



O

- g este cel mult f
- f ∈ O(f)

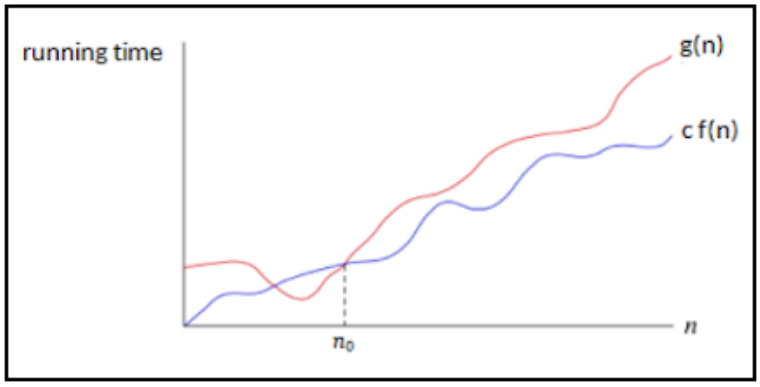
$$O(f(n)) = \{g : \mathbb{R}^+ \to \mathbb{R}^+ \mid \begin{matrix}\exists c \in \mathbb{R}^+ \\ \exists n_0 \in \mathbb{N}\end{matrix} \; such\ that \; \forall\, n \geq n_0,\ \ 0 \leq g(n) \leq c f(n)\}$$



Ω

- g este cel putin f
- f ∈ Ω(f)

$$\Omega(f(n)) = \{g : \mathbb{R}^+ \to \mathbb{R}^+ \mid \begin{matrix} \exists c \in \mathbb{R}^+ \\ \exists n_0 \in \mathbb{N} \end{matrix} \; such\ that\ \forall\ n \geq n_0,\ \ 0 \leq cf(n) \leq g(n)\}$$



o

- g este STRICT mai lent decat f
- o(f(n)) = O(f(n)) \ Θ(f(n))
- f ∉ o(f)

$$o(f(n)) = \{g : \mathbb{R}^+ \to \mathbb{R}^+ \mid \begin{matrix} \forall c \in \mathbb{R}^+ \\ \exists n_0 \in \mathbb{N} \end{matrix} \; such\ that\ \forall\ n \geq n_0,\ \ 0 \leq g(n) \leq cf(n)\}$$

ω

- g este STRICT mai rapid decat f
- w(f(n)) = Ω(f(n)) \ Θ(f(n))
- ω ∉ o(f)

$$\omega(f(n)) = \{g : \mathbb{R}^+ \to \mathbb{R}^+ \mid \begin{matrix} \forall c \in \mathbb{R}^+ \\ \exists n_0 \in \mathbb{N} \end{matrix} \; such\ that\ \forall\ n \geq n_0,\ \ 0 \leq cf(n) \leq g(n)\}$$

Exercitii

Exercitiul 1: Dați exemple de câte o funcție din următoarele clase de complexitate:

Fie a,b,c,p numere reale

O(n):

- f(n) = 3n + 5
  Demonstratie: Gasim c a.i. $0 \le f(n) \le c * n$
  $c = 4$ a.i. $0 \le 3n + 5 \le 4 * n$ pentru $\forall\ n \ge 5$
- f(n) = a * n + b
- f(n) = log(n)
- f(n) = c
- f(n) = a * n + sin(n)
  Demonstratie: $-1 \le \sin(n) \le 1$
  $\Rightarrow c_1 = a - 1$; $c_2 = a + 1$ a.i. inegalitatea se verifica pentru $\forall\ n \ge 1$
  $0 \le (a - 1) * n \le a * n + \sin(n) \le (a + 1) * n$

Ω(log(n)):

- f(n) = n ^ p
- f(n) = a * n + b
- f(n) = c * log(n)

Θ(n^2):

- f(n) = a * n^2 + b * n + c
- f(n) = a * n^2 + log(n)
- f(n) = a * n^2 + sin(n)
  Demonstratie: Gasim c_1, c_2 a.i.
  $c_1 * n^2 \le a * n^2 + \sin(n) \le c_2 * n^2$
  $c_1 = a - 1$; $c_2 = a + 1 \Rightarrow$ inegalitatea se verifica pentru $\forall\ n \ge n0$

ω(1/n):

- f(n) = n^p , $\forall\ p \ge 0$
- f(n) = p^n, $\forall\ p \ge 1$
- f(n) = log(n)

o(3^n):

- f(n) = c^n, $0 \le c < 3$
- f(n) = log(n)
- f(n) = $a * n^p + b * n^{(p-1)} + ... + k_0 * n^0$
  Observatii: polinomial ≤ exponential $\forall\ n \ge n0, p \ge 1$
  $a_1*n^p + b_1 * n^{p-1} + ... + k_1 * n^0 \le a_2 * q^n + b_2 * q^{n-1} + ... + k_2 * q^0$ $\forall\ n \ge n0, p \ge 1$

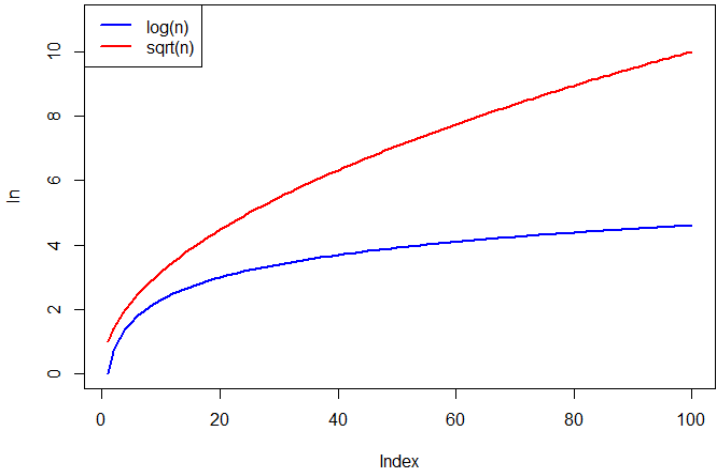
Exercitiul 2: Pentru fiecare din perechile de mai jos, dați un exemplu concret de constante pentru care inegalitățile implicate în definiția clasei de complexitate sunt adevărate:

- **$\log(n) \in o(\sqrt{n})$**

  $\forall\ c \in R\ ,\ \exists\ n_0 \in N$ astfel incat

  $\forall\ n \geq n_0,\ 0 \leq g(n) \leq c * f(n)$

  $0 \leq \log(n) \leq c * \sqrt{n}$



  Putem alege c a.i. relatia sa se verifice pentru $\forall\ n \geq n_0$, unde alegem n_0=1024:

  $0 \leq \log(n) \leq c * \sqrt{n}$

  $c \leq \sqrt{n} / \log(n)$

  Acest lucru se intampla pentru $\forall$ c, de la un anumit prag.

- **$100 * n^2 \in O(2^n)$**

  $\exists\ c \in R, \exists\ n_0 \in N$ a.i. $0 \leq g(n) \leq c * f(n)\ \ \forall\ n \geq n_0,$

  $100 * n^2 \leq c * 2^n$

  Alegem $n_0 = 1 \Rightarrow 100 \leq c * 2 \Rightarrow c = 50$

- **$0.001 * n^3 \in \Omega(20 * \sqrt{n})$**

  $\exists\ c \in R, \exists\ n_0 \in N$ a.i. $0 \leq c * 20 * \sqrt{n} \leq 0.001 * n^3\ \ \forall\ n \geq n_0,$

  Alegem $n_0 = 1 \Rightarrow c * 20 \leq 0.001 \Rightarrow c = 1/(2*10^4)$

- **$10n \in \Theta(n)$**

  $\exists\ c_1, c_2 \in R, \exists\ n_0 \in N$ a.i. $c_1 * n \leq 10 * n \leq c_2 * n\ \ \forall\ n \geq n_0,$

  $c_1 = 9; c_2 = 11$

- **$n \in \Theta(10 * n)$**

  $\exists\ c_1, c_2 \in R, \exists\ n_0 \in N$ a.i. $c_1 * 10 * n \leq n \leq c_2 * 10 * n\ \ \forall\ n \geq n_0,$

  $c_1 = 1/20, c_2 = 1/5$

Exercitiul 3: Gasiti f,g a.i. sa se indeplineasca afirmatiile:

- **$f(n) = O(g^2(n))$**

  !!! Aceasta notatie cu "=" este un Syntactic sugar

  "$\exists$ o functie $h(n) \in O(g^2(n))$ a.i. $f(n) = h(n)$ $\forall$ n"

  $f(n) \in O(g^2(n))$

  Alegem $g(n) = \sqrt{n} \Rightarrow$ Gasim $f(n) \in O(n) \Rightarrow f(n) = n$

- **$f(n) = \omega(\log(g(n)))$**

  f(n) sa fie strict mai rapida ca log(g(n))

  Alegem $f(n) = n^2$ si $g(n) = n$

- **$f(n) = \Omega(f(n) * g(n))$**

  f(n) sa fie mai rapid (sau egal ca rata de crestere) cu f(n) * g(n)

  Alegem $f(n) = n$, $g(n) = 1/n$

- **$f(n) = \Theta(g(n)) + \Omega(g^2(n))$**

  Alegem $g(n) = 1$

  $\Theta(g(n)) + \Omega(g^2(n)) = \Theta(1) + \Omega(1) = \Omega(1)$ (scriere posibila datorita syntactic sugars)

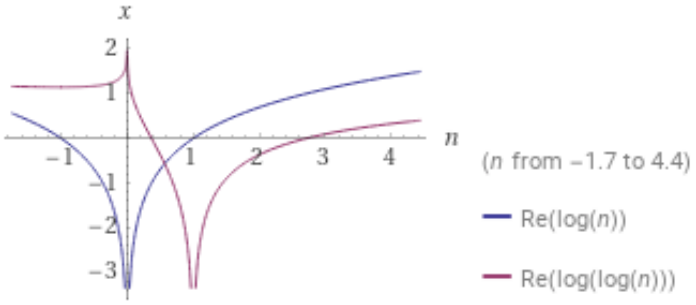
  f(n) in $\Omega(1) \Leftrightarrow$ f mai rapida sau cu aceeasi rata de crestere ca 1

  Deci putem alege $f(n) = n$

Exercitiul 4: Verificați valoarea de adevăr a următoarelor propoziții:

- **$\sqrt{n} \in O(\log(n))$** $\sqrt{n} > \log(n)$ $\forall$ $n \geq 0$ → **NU**

- **$\log(n) \in O(\log(\log(n)))$** → **NU** (vezi grafice)



- **$n \in O(\log(n) * \sqrt{n})$** $\sqrt{n} > \log(n) \Rightarrow n > \sqrt{n} * \log(n)$ $\forall$ $n \geq 0$ → **NU**

- **$n + \log(n) \in \Theta(n)$** $n \leq n + \log(n) \leq n * 2$ → **DA**

  $\Rightarrow$ Se pot alege valorile $c_1 = 1$, $c_2 = 2$ pentru inegalitatea din definitia lui Θ

- **$\log(n * \log(n)) \in \Theta(\log(n))$**

  $\log(n * \log(n)) = \log(n) + \log(\log(n))$

  $\log(n) \leq \log(n) + \log(\log(n)) \leq 2 * \log(n)$ $\forall$ $n \geq 5$ → **DA**

- **$\sqrt{n} \in \omega(\log(n))$** → **DA**

Syntactic sugars

Notatiile asimtotice sunt adesea folosite pentru a ne referi la functii arbitrare cu o anumita crestere, Pentru simplitate, putem scrie *expresii aritmetice* astfel:

$$f(n) = \Theta(n) + O(\log(n))$$

care trebuie citita:

$$\exists g \in \Theta(n) \text{ și } \exists h \in O(\log(n)) \text{ a.î. } f(n) = g(n) + h(n),\ \forall n \in \mathbb{R}^+ .$$

Putem de asemenea scrie *ecuatii*:

$$\Theta(n^2) = O(n^2) + o(n)$$

care ar trebui citite:

$$\forall f \in \Theta(n^2),\ \exists g \in O(n^2) \text{ și } \exists h \in o(n) \text{ a.î. } f(n) = g(n) + h(n),\ \forall n \in \mathbb{R}^+ .$$

Observati ca ecuatiile nu sunt simetrice si ar trebui citite de la stanga la dreapta. De exemplu:

$$\Theta(n) = O(n)$$

Desi e adevarat ca, pentru orice functie $\Theta(n)$ exista o functie egala in $O(n)$, putem vedea ca exista functii in $O(n)$ pentru care nu exista niciun corespondent in $\Theta(n)$.

Ca regula, fiecare notatie asimtotica din stanga unui egal, ar trebui citita ca o functie **cuantificata universal** ($\forall$ f) din aceea clasa, iar fiecare notatie asimtotica din dreapta unui egal ar trebui citita ca o functie **cuantificata existential** ($\exists$ g) din clasa respectiva.

$$\left(\frac{\omega(n^2)}{\Theta(n)}\right) = \Omega(n) + o(n)$$

$$\forall f \in \omega(n^2)\ and\ \forall g \in \Theta(n),\ \exists h \in \Omega(n) \text{ și } \exists j \in o(n) \text{ a.î. } \left(\frac{f(n)}{g(n)}\right) = h(n) + j(n),\ \forall n \in \mathbb{R}^+$$

(egalitate adevarata doar in directia stanga → dreapta)

$\Theta(n^2) = O(n^2) + o(n)$

Orice functie f ∈ $\Theta(n^2)$ se poate scrie ca o suma de $f_1 \in O(n^2)$ si $f_2 \in o(n)$

ex: f(n) = n^2 + sin(n)

f_1(n) = n^2

f_2(n) = sinn

ex: f(n) = n^2 +16

f_1(n) = n^2

f_2(n) = 16

dreapta → stanga : egalitatea NU tine

f_1(n)= log(n)

f_2(n)= log(n)

f(n) = 2* log(n) ∉ $\Theta(n^2)$

Θ(n) = Ω(n)

⇔ " Pentru orice f ∈ $\Theta(n)$, exista o functie g ∈ Ω(n) a.i. f(n) = g(n) ∀ n"

stanga → dreapta: f(n) = c * n ∈ Θ(n) ⇒ g(n) = c * n ∈ Ω(n)

dreapta → stanga: Putem sa gasim functii din O(n) sau Ω(n) a.i sa nu fie egale cu o functie din Θ(n)

Θ(n) = O(n)

⇔ "Pentru ∀ f ∈ Θ(n), exista o functie g ∈ O(n) a.i . f(n) = g(n) ∀ n"

Exercitiul 5: Scrieti sub forma mai simpla:

- **O(n√n) / Θ(n) = O(√n)**

  "Orice f ∈ O(n * √n) si g ∈ Θ(n), raportul f/g se poate scrie ca o functie h ∈O(√n)"

  SAU

  "∃ h ∈ O(sqrt(n)) a.i h = f/g"

  Rezolvare:

  Fie f(n) ∈ O(n * √n) si g(n) ∈ Θ(n)

  ∃ $c_1, c_2 > 0$:

  $c_1 * n \leq g(n) \leq c_2 * n$

  $1/(c_1 * n) \geq 1/g(n) \geq 1/(c_2 * n)$

  $1/(c_2 * n) \leq 1/g(n) \leq 1/(c_1 * n)$

  ∃ c > 0:

  $0 \leq f(n) \leq c * n * \sqrt{n}$

  $0 \leq f(n)/g(n) \leq c * n * \sqrt{n} / (c_1 * n) = (c/c_1) * \sqrt{n}$

  $0 \leq f(n)/g(n) \leq (c / c_1) * \sqrt{n}$

  ⇒ ∃ $c_3 = (c/c_1)$ care indeplineste inegalitatea din stanga = definitia lui O(√n)

- **Θ(n) / O(log(n)) = Ω(n/log(n))**

  ∃ $c_1, c_2 > 0$:

  $c_1 * n \leq f(n) \leq c_2 * n$

  ∃ c > 0:

  $0 \leq f(n) \leq c * \log(n)$

  $+\infty \geq 1 / g(n) \geq 1 / (c * \log(n))$

  Dar $c_2 * n \geq f(n) \geq c_1 * n$

  $+\infty \geq f(n) / g(n) \geq c_1 * n / (c * \log(n)) = (c_1/c) * (n/\log(n))$

  Cum $c_1/c > 0$ (notat c_3) si $n/\log(n) > 0$ de la un anume prag

  ⇒ $f(n) / g(n) \geq c_3 * (n/\log(n)) > 0$

  ⇒ inegalitatea este definitia lui Ω(n/logn)

Exercitiul 6: Demonstrați/ infirmați următoarele propoziții:

- **f(n) = Ω(log(n))**
  g(n) = O(n)
  ⇒ **f(n) = Ω(log(g(n))**

  DA

  $\exists\ c_1 > 0$: $0 \le c_1 * \log(n) \le f(n)$

  $\exists\ c_2 > 0$: $0 \le g(n) \le c_2 * n$

  $\Rightarrow \log(g(n)) \le \log(c_2) + \log(n)$

  $\Rightarrow \log(g(n)) - \log(c_2) \le \log(n)$

  $\Rightarrow \log(g(n)) - k \le \log(n)$

  $\Rightarrow f(n) \ge \log(n) * c_1 \ge c_1 * (\log(g(n)) - k)$

  $= c_1 * \log(g(n)) - c_1 * k$

  $\ge c_3 * \log(g(n))$ unde c_3 e o constanta > 0

  $c_3, c_2, c_1, k > 0$:

  $f(n) \ge c_3 * \log(g(n)) \ge 0$ deci definitia lui $\Omega(\log(g(n))$

- **f(n) = Ω(log(n))**
  g(n) = O(n)
  ⇒ **f(n) = Θ(log(g(n))**

  NU

  Am obtinut la exercitiul anterior $\Omega(\log(g(n))$, dar nu avem o margine superioara.
  Si cum $\Omega(\log(g(n)) \neq \Theta(\log(g(n))$

  (de ce: Putem gasi g(n) care $\in \Omega(\log(g(n))$ si $\notin \Theta(\log(g(n)))$

  $\Rightarrow f(n) \neq \Theta(\log(g(n))$

- **f(n) = Ω(g(n))**
  g(n) = O(n^2)
  ⇒ **g(n) / f(n) = O(n)**

  DA

  $\exists\ c_1 > 0$: $0 \le g(n) \le c_1 * n^2$

  $\exists\ c_2 > 0$: $0 \le c_2 * g(n) \le f(n) \le inf+$

  $0 \le 1 / f(n) \le 1 / (c_2 * g(n))$

  $\Rightarrow 0 \le g(n) / f(n) \le g(n) / (c_2 * g(n))$

  $\Rightarrow 0 \le g(n) / f(n) \le c_2$

  Am ajuns la definitia lui O(1)

  ⇒ g(n) / f(n) = O(1) = (Syntactic sugars) = O(n)

  pentru ca pentru orice $f(n) \in O(1)$, $\exists\ h(n) \in O(n)$ a.i. $f(n) = h(n)$